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PROFESSIONAL SUMMARY

- 7.6+ years of professional experience in **Generative AI, Agentic AI, Machine Learning, MLOps, LLMs, RAG, NLP, AWS Bedrock, Python, PySpark, SQL, Databricks, CI/CD Pipelines, Kubernetes**, and major cloud platforms (**AWS, Azure, GCP**).
- Proficient in **GenAI tools** like **LangChain, LangGraph, OpenAI models**, and **RAG-based architectures** with hands-on experience in building, fine-tuning, and deploying **LLM** pipelines using **AWS SageMaker, Azure ML**, and **GCP Vertex AI**.
- Designed and implemented **data models** and **data warehouses** using tools like **Azure Data Factory, AWS Glue**.
- **Optimized data storage and processing performance** using big data frameworks (e.g., Spark, Databricks, Hadoop) and cloud-native storage solutions (e.g., S3, Azure Data Lake).
- Developed and deployed Appian applications using **ADK**, including process models, SAIL interfaces, and custom plug-ins to meet business requirements.
- Integrated external systems and automated workflows via **ADK**, enabling secure data exchange, faster deployments, and improved process efficiency.
- Strong experience with AI/ML frameworks including **TensorFlow, PyTorch, Scikit-learn, XGBoost, LightGBM**, and classical model tuning to optimize **MAE/RMSE/R²** scores with A/B testing and cross-validation strategies.
- **API Deployment** for Deploy APIs to **handle client requests** reliably, ensure **scalable and highly available** endpoints and implement **security**: authentication, authorization, and TLS
- Experience in **Natural Language Processing (NLP)** including **tokenization, NER, POS tagging, sentiment analysis**, text generation, and **speech-to-text** processing to power **Conversational AI** systems.
- Developed and optimized text to speech models that achieved human like voice synthesis, maintaining the unique style of voice actors across multiple languages.
- Skilled in designing, deploying, and managing production-grade ML/AI workflows using **Amazon SageMaker, Google Vertex AI, and Azure Machine Learning**, with experience in **automated pipelines, model versioning, monitoring, rollback, and retraining**.

- Hands-on with **vector databases** like **Pinecone, ChromaDB, FAISS**, and integration with GenAI-based search and retrieval systems.
- Proficient in **MLOps tools and practices** including **MLflow, Kubeflow, Airflow, DVC**, and containerization/orchestration using **Docker** and **Kubernetes**. Built and maintained CI/CD pipelines with **GitHub Actions, Jenkins, AWS CodePipeline**, and Azure DevOps.
- Broad experience with **AWS services**: EC2, Lambda, IAM, S3, CloudFormation, ECS, EKS, RDS, Route 53, SNS, SES, SageMaker, and **Amazon Bedrock** for LLM integration.
- Proficient in **Azure Cognitive Services** (Vision, Speech, Language, Decision, Search), **Azure OpenAI**, and **Azure Databricks** for scalable AI model deployment.
- Experienced **Data Engineer** with expertise in designing and optimizing **ETL/ELT pipelines**, batch and real-time processing using **PySpark, SQL, and Python** on **cloud-native GCP services** such as **BigQuery, Dataflow, and Dataproc**.

EDUCATION:

- Master of Technology [M. Tech], RGM CET.

EXPERIENCE:

- Worked at **Incedo** as an AI/ML Engineer in Bangalore from July 2024 to May 2025.
- Worked at **iSpace Software Solutions** as an AI/ML Engineer in Bangalore from January 2023 to June 2024.
- Worked at **Sterlite Technologies** (merged with **STL Digital Limited**) as an ML Engineer in Bangalore from May 2022 to January 2023.
- Worked at **HCL Technologies** as an MLOps Engineer in Bangalore from October 2018 to April 2022.
- Worked at **HARMAN** as a DevOps Engineer in Bangalore from November 2017 to June 2018.

EXPERIENCE

Incedo

Responsibilities:

- Designed and developed scalable data ingestion pipelines to securely stream patient PDFs from Azure Blob Storage using BlobServiceClient, eliminating local storage overhead.
- Implemented high-accuracy PDF text extraction with PyMuPDF, preserving layout and supporting scanned documents, enabling clean and structured input for downstream NLP tasks.

- Engineered context-aware text chunking using **LangChain's RecursiveCharacterTextSplitter**, applying hierarchical splitting with token overlap to optimize **LLM input** windows.
- Generated semantic embeddings with **OpenAI's text-embedding-ada-002** (dim vectors) and enriched them with metadata for contextual search.
- Deployed high-performance semantic search infrastructure by indexing embeddings in Pinecone, achieving sub-100ms retrieval with **cosine similarity** across millions of chunks.
- Built retriever modules (**RAG pattern**) leveraging Pinecone + LangChain, enabling top-k similarity search and filtering of relevant context for user queries.
- Integrated **GPT-4** with **LangChain RetrievalQA**, orchestrating query → retrieval → response pipelines to generate grounded, context-aware, and citation-backed answers.
- Deployed end-to-end **GenAI solution** on Azure (Azure OpenAI, Document Intelligence), integrating with web UI (Streamlit/Flask) for interactive query-response and traceable document intelligence.
- Integrated external systems and automated workflows via **ADK**, enabling secure data exchange, faster deployments, and improved process efficiency.
- Developed and optimized text to speech models that achieved human like voice synthesis, maintaining the unique style of voice actors across multiple languages.
- Implemented real time processing solutions that reduced inference time to under 1 second, enhancing user interaction and experience.

iSpace Software Solution

Responsibilities:

- Designed and architected scalable **NLP solutions** aligned with business requirements.
- Documented solution architecture, design decisions, and implementation steps to support cross-functional collaboration.
- Built and optimized computer vision models for tasks like **image classification, object detection, segmentation, and tracking**.
- Trained, **fine-tuned, and optimized ML/DL models** for real-time NLP and CV use cases.
- Applied strong proficiency in Python, machine learning algorithms, NLP pipelines, and deep learning frameworks.
- Developed Generative AI solutions using **GPT models, LangChain, and LLMs** with **embedding-based search and retrieval**.
- Implemented LLM-driven use cases for **classification, extraction, summarization, semantic search, and conversational AI** on unstructured text data.
- Developed scalable document loaders to ingest data from Azure Blob Storage and load into vector databases.
- **Fine-tuned pre-trained Word2Vec embeddings** on domain-specific data to enhance semantic understanding and performance.
- Deployed and optimized **ChromaDB-based vector** stores for high-speed and accurate similarity search across large datasets.
- Designed and implemented custom retrieval algorithms for fetching relevant vectors/documents based on user queries.

- Built and deployed end-to-end **MLOps pipelines** and cloud-native AI solutions on Microsoft Azure.
- Developed and deployed Appian applications using **ADK**, including process models, SAIL interfaces, and custom plug-ins to meet business requirements.
- Managed large scale datasets for voice cloning project and ensuring high performance and reliability while supporting multilingual transcription.

STL Digital (Sterlite Technologies)

Responsibilities:

- Designed and developed Generative AI models and algorithms using advanced techniques such as **GPT and GANs**.
- Built rapid Proof of Concept (PoC) projects to validate AI-driven business solutions and showcase feasibility.
- Proficient in **Python**, with hands-on experience in **TensorFlow, Keras, and other ML/DL frameworks**.
- Utilized Python libraries like **Pandas, Scikit-learn, NumPy, NLTK, spaCy, and SciPy** for data preprocessing, modeling, and evaluation.
- Developed end-to-end applications leveraging **Generative AI, NLP, LLMs, GPT** models, and deep learning techniques.
- Deployed machine learning models into production environments on cloud platforms such as Azure.
- Managed **ML model** lifecycle including training, retraining, versioning, and performance monitoring using **MLOps** open-source frameworks.
- Designed and implemented cloud-native ML solutions and **MLOps pipelines on Azure Cloud**.
- Automated and orchestrated the **CI/CD process** for seamless model deployment and production monitoring.

HCL Technologies

Responsibilities:

- Oversaw the utilization of **AWS SageMaker** for end-to-end machine learning workflows, including **model training, deployment, and monitoring**.
- Designed and developed **end-to-end ETL pipelines** using **PySpark** on **Azure Databricks** to ingest, cleanse, and transform large-scale structured and semi-structured data from multiple sources into **Azure Data Lake** and **SQL-based data warehouses**.
- Implemented scalable **data processing workflows** in **Azure Databricks**, leveraging **Delta Lake, PySpark, and ADF** to **automate data integration**, improve pipeline performance, and support real-time analytics and reporting needs.
- Proficient in **Python**, with hands-on experience in machine learning libraries and deep learning frameworks such as **TensorFlow, Scikit-learn, and Keras**.
- Designed and implemented cloud-native ML solutions and built scalable **MLOps pipelines** on AWS Cloud.

- Managed infrastructure requirements for ML and data science projects, including provisioning and optimizing **AWS resources** and **SageMaker instances**.
- Drove innovation in **ML workflows** by evaluating and integrating emerging tools and techniques to improve model accuracy and efficiency.
- Designed, deployed, and maintained scalable and reliable infrastructure for production-grade ML model deployment and operations.
- Automated and orchestrated the **CI/CD pipelines** for seamless ML lifecycle management and version-controlled deployments.
- Collaborated with data scientists to deploy models, monitor their performance, and resolve issues related to scalability, accuracy, and stability in production.

Harman Connected Services Corp..

Responsibilities:

- Designed and implemented **Jenkins multi-branch CI/CD pipelines** using **Groovy**, automating the build, test, and deployment processes for various applications.
- Created and maintained **Docker images** for consistent and portable application deployments across environments.
- Managed and deployed **containerized applications** using **Kubernetes**, handling **pods, deployments, services, config maps, and Helm charts**.
- Developed **Python scripts** for automation, monitoring, deployment orchestration, and system health checks.
- Configured and managed **Git repositories**, branches, pull requests, and webhooks to ensure smooth version control and collaboration.
- Integrated **SonarQube** with CI pipelines for static code analysis and enforced **quality gates** to maintain code standards.
- Used **JaCoCo** and **JUnit** for code coverage and unit testing, ensuring software quality in automated pipelines.
- Automated **build and packaging** workflows using **Maven**, and integrated continuous deployment to various environments.
- Implemented **infrastructure as code (IaC)** using **Ansible** for provisioning and configuration management.
- Reviewed and validated **Ansible playbooks and roles** for security, compliance, and operational consistency.
- Monitored and troubleshooted pipeline executions, deployment issues, and performance bottlenecks across CI/CD systems.
- Designed and managed **CI/CD automation workflows** integrated with **Docker, Kubernetes, Jenkins, and cloud platforms (Azure)**.